Thirumalainambi M

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Summary

AI Engineer experienced in building production-grade ML systems, scalable LLM applications, and cloud-native AI on AWS. Currently at Alyson.ai working on end-to-end ML pipelines, MLOps, and distributed systems. Strong problem-solving background (Codeforces Specialist, 4 CodeChef, top 4 percent LeetCode with 1000+ day streak). Passionate about scalable, high-performance AI systems.

Education

Sinhgad College of Engineering, Pune June 2025
*B.E. Electronics & Telecommunication – **DISTINCTION***

Lakshya Prep Junior College, Mumbai 2021
HSC – 92.50%

The South Indian Association High School, Mumbai 2019
SSC – 90.80%

Technical Skills

Programming: Python, Java, TypeScript, SQL
AI/ML & LLM Engineering: PyTorch, TensorFlow, Scikit-learn, LLM , OpenClaw, Prompt Engineering, RAG, Embeddings, Vector Databases, Model Evaluation
AI Systems & MLOps: Model Deployment, CI/CD for ML, ML Pipelines, Model Versioning, Monitoring, A/B Testing, Docker, Kubernetes
Cloud Infrastructure: AWS (S3, EC2, Lambda, SageMaker, Athena, CloudWatch), Serverless Architectures, Distributed Systems
Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, Microservices Architecture
Databases: PostgreSQL, MySQL, MongoDB, Firebase, Vector Databases
Core CS: System Design, Distributed Systems, Operating Systems, DBMS, Computer Networks, OOPs
Tools: Git, GitHub, GitHub Actions, Cursor, Claude Docker, Kubernetes, Postman, n8n

Professional Experience

Alyson.ai, Remote – San Francisco, California Aug 2025 – Present
AI Engineer / Data Scientist

- Built and curated **10+ production datasets**, improving **data integrity by 30%** and transforming raw data into unified analytical layers using **Medallion Architecture** (Bronze, Silver, Gold).
- Designed and deployed scalable **ML pipelines** powering production AI systems with automated training, evaluation, deployment, and **CI/CD integration**.
- Architected cloud-native ML infrastructure using **AWS** (**S3**, EC2, Lambda, SageMaker, CloudWatch) with monitoring, logging, and **model versioning** for reliable production performance.
- Optimized model inference latency and system throughput for **real-time AI applications**, enabling low-latency predictions in production.
- Built an **Agentic AI HR Platform** that automated employee **onboarding, offboarding, leave management, bonus management, and time tracking**, significantly reducing manual HR operations and streamlining payroll workflows.
- Developed an AI-powered **Meeting Intelligence Platform** using **Recall.ai**, where bots automatically joined meetings, captured real-time transcripts, stored meeting data in **Amazon S3**, and generated reporting, analytics, and cost management dashboards.
- Designed a personalized **Knowledge Brain** for every employee by indexing meeting transcripts and organizational knowledge, enabling context-aware AI assistants for semantic search, historical meeting retrieval, and organization-wide knowledge discovery.

LeetCode: pabloescobar69
CodeChef: badge
GeeksforGeeks: thiru2905
Codeforces: thiru2905

Technical Projects

DumpStash AI | *Production-ready AI knowledge engine integrating LLM workflows*

- Designed **RAG-based architecture** using embeddings and vector retrieval.
- Built scalable backend services with authentication, storage, and **AI orchestration**.
- Optimized prompt pipelines and inference workflows for performance and reliability.

Collaborative System Design Playground | *Real-time collaborative platform for system design*

- Built real-time synchronization using **WebSockets**.
- Designed scalable backend infrastructure for **concurrent users**.
- Applied distributed systems principles for scalability and consistency.

Achievements

Competitive Programming & Recognition

- Achieved **1400 (Specialist)** rating on Codeforces.
- Secured **4-star** rating on CodeChef.
- Ranked in the **top 2%** globally (rating 1983) on GeeksforGeeks.
- Ranked in the **top 4.7%** globally (rating 1890) on LeetCode.
- Won **Project of the Month (Oct 2024)** on Peerlist.
- Maintained a **1000-day+** solving streak on LeetCode.
- Freelanced for clients from **France and Germany** on SaaS projects.

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